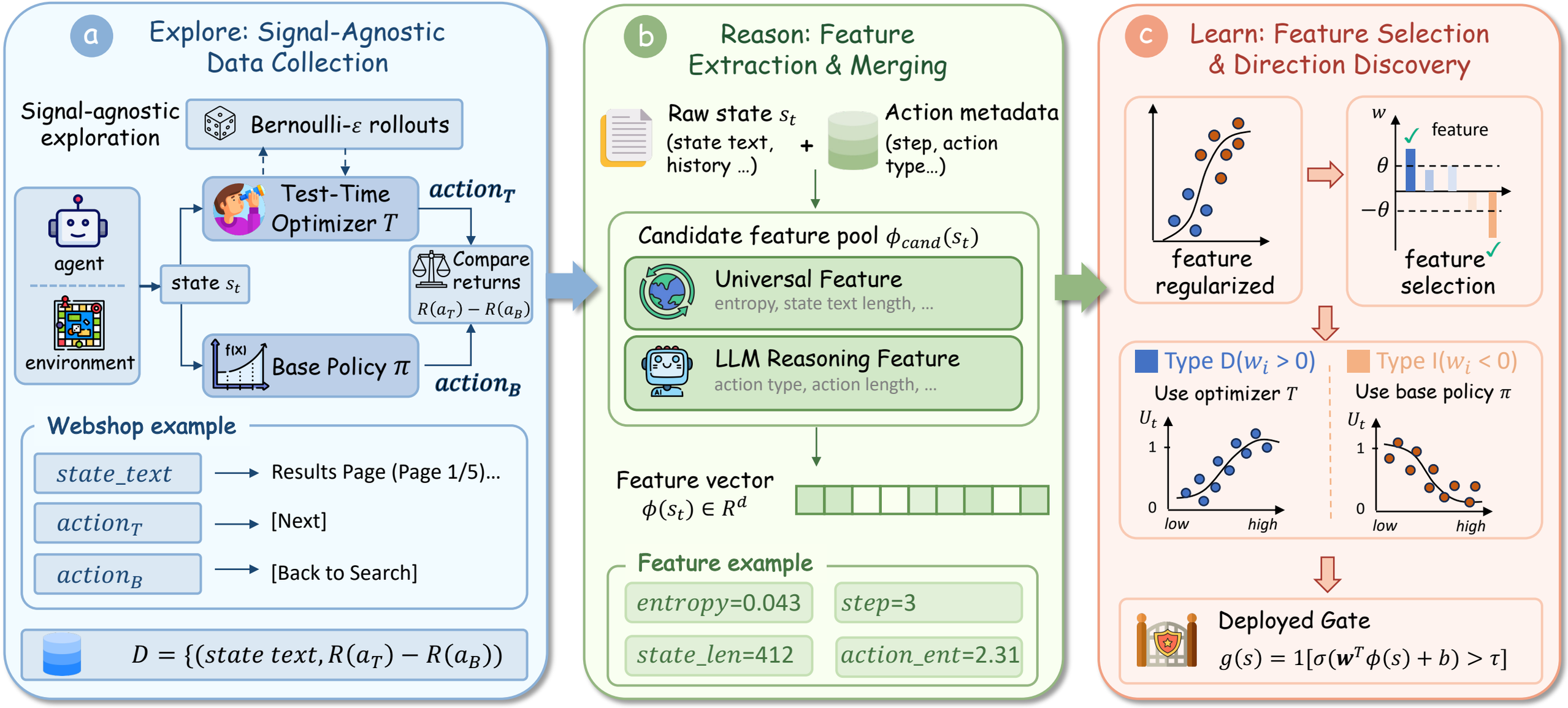
a
Explore: Signal-Agnostic Data Collection
Signal-agnostic exploration
Bernoulli-ε rollouts
Test-Time Optimizer T
action_T
agent
environment
state s_t
Compare returns
R(a_T) − R(a_B)
f(x)
Base Policy π
action_B
Webshop example
state_text
Results Page (Page 1/5)...
action_T
[Next]
action_B
[Back to Search]
D = {(state text, R(a_T) − R(a_B))
b
Reason: Feature Extraction & Merging
Raw state s_t
(state text, history ...)
+
Action metadata
(step, action type...)
Candidate feature pool φ_cand(s_t)
Universal Feature
entropy, state text length, ...
LLM Reasoning Feature
action type, action length, ...
Feature vector
φ(s_t) ∈ R^d
Feature example
entropy=0.043
step=3
state_len=412
action_ent=2.31
c
Learn: Feature Selection & Direction Discovery
feature regularized
w
feature
θ
−θ
feature selection
Type D(w_i > 0)
Use optimizer T
Type I(w_i < 0)
Use base policy π
U_t
1
0
low
high
Deployed Gate
g(s) = 1[σ(w^T φ(s) + b) > τ]